Motion Graphs Teacher Packet

Consolidated teacher-use materials, assessment tools, and reflection notes for the toy car motion-graphs lesson

Packet purpose

This packet brings the updated teacher-facing resources into one place. It includes the revised assessment recording sheet, planning reflections, and the student and teacher handouts needed to run the lesson smoothly.

Lesson focus	Students investigate one-dimensional motion using a toy car and connect position-time, velocity-time, and acceleration-time graphs through slope, area, and graph shape.
Main evidence sources	Teacher observation and conferencing, student booklet, group poster, and 3-2-1 connect and reflect sheet.
Profiles centered in planning	Michael (written output and assistive technology), Anish (processing and organization), Sandra (MLL language support), with accessibility notes also built in for Juan.

What is included in this packet

Use the pages that follow as a print-and-go teacher packet. The merged packet includes the revised teacher assessment sheet plus the updated student and teacher handouts.

- Revised teacher assessment recording sheet that matches the live observation, booklet, poster, and 3-2-1 evidence sources used in the lesson.
- Student booklet for the toy car investigation and graph conversions.
- Investigation checklist for group pacing and teacher checkpoints.
- Vocabulary support sheet and scaffolded graph templates for accommodations and differentiation.
- Group role cards and Slo-Mo quick guide for smoother lab setup and clearer routines.
- 3-2-1 connect and reflect sheet for self-assessment, peer feedback, and metacognition.

Teacher prep notes

- Print the student booklet double-sided if possible. Keep extra graph paper and scaffolded graph templates available rather than giving them to every student immediately.
- Cut the group role cards before class and assign or guide roles intentionally so writing-heavy and physically demanding tasks can be redistributed when needed.
- Project the slide deck and keep the Slo-Mo quick guide visible during setup so groups can self-correct before filming.
- Use the revised assessment sheet during circulation. Record live evidence first, then add product evidence after collecting the booklet, poster, and 3-2-1 sheets.

Assessment coding

Live observation columns use 2 = secure, 1 = developing, 0 = with prompt, and N = not observed. Product columns use the same scale so the teacher can see at a glance whether understanding was shown during action, after collection, or both.

Assessment at a glance

This lesson uses written, oral, and performance evidence. Students do not rely on one single product. Understanding can appear in real-time talk, graphing moves, numerical work, the shared poster, and the final reflection.

Evidence source	What students do	What the teacher records
Observation / conferencing	Set up the track, collect data, explain slope and sign, convert graphs, and justify what shape makes sense.	A1-A5, oral communication, collaboration, and quick conference notes during the action portion.
Student booklet	Complete tables, graphs, slope work, area/displacement interpretation, and brief written responses.	Accuracy of graph setup, calculations, conversion logic, and cross-checking of displacement.
Group poster	Present aligned p-t, v-t, and a-t graphs with labels, time window, and concise graph-based claims.	Whether the group product shows clear graph alignment, correct labels, and accurate graph statements.
3-2-1 connect and reflect	Make connections, review peer work, and reflect on what the graphs mean.	Metacognitive understanding, peer feedback quality, and remaining misconceptions to address next class.

Why the revised recording sheet is stronger

The original checklist captured important live observations well, but the updated version now mirrors the full activity more closely by leaving room for observation evidence and later product evidence. This makes it easier to see whether a student understood the idea during the lesson, on paper after the lesson, or in both places.

Teacher Assessment Recording Sheet - Motion Graphs

Live observation: A1-A5, Comm, Collab | Product evidence: B = Booklet, P = Poster, R = 3-2-1

Codes: 2 = secure | 1 = developing | 0 = with prompt | N = not observed	A1 axes + units | A2 best-fit p-t | A3 p-t to v-t | A4 v-t to a-t | A5 area on v-t = displacement

Student	A1	A2	A3	A4	A5	Comm	Collab	B	P	R	Notes / next step
1.											
2.											
3.											
4.											
5.											
6.											
7.											
8.											
9.											
10.											
11.											
12.											
13.											

Whole-class next steps for the next lesson or next block:	**Students to conference or check first next time:**

Differentiation, accommodations, and modifications

This lesson is differentiated through content access, process supports, product options, and classroom environment. No curriculum modifications are planned for the selected learner profiles. All students are working toward the same graph interpretation and graph conversion expectations, but they are not expected to access or communicate that learning in exactly the same way.

> **Selected profiles for rationale**
>
> The explanation below focuses on Michael, Anish, and Sandra because they show three different types of support planning in this lesson: written output and assistive technology, processing and organization, and language access. Accessibility notes for Juan are still built into the packet through flexible roles, accessible setup, and seating considerations.

Michael - written output and assistive technology

Michael's main barrier in this lesson is not understanding the motion ideas themselves; it is the physical and written-output demand created by repeated graphing, copying, and reflection writing. To address this, the packet includes a structured booklet, digital copies, scaffolded graph templates, and flexible response options such as point form, typed work, or oral conferencing. These choices were planned so that assessment stays focused on axes, units, slope, area, and graph interpretation rather than on handwriting speed or mechanics. This is an appropriate accommodation because the task still asks him to reason through the same p-t, v-t, and a-t relationships as the rest of the class.

Anish - processing and organization

Anish benefits from clear visual structure, chunked instructions, and predictable routines. The investigation checklist, visible task sequence, guided role cards, and teacher checkpoint structure were chosen so that each phase of the lesson is easier to follow before the group moves on. These supports reduce the chance that he loses track of which graph comes next or what evidence is needed at each step. The rationale is that the lesson includes several linked stages, and organizational supports help him show what he knows without lowering the conceptual demand of the graphing work.

Sandra - MLL language and vocabulary access

Sandra needs support with the language load of the lesson rather than with the core physics ideas alone. For that reason, the packet includes a vocabulary support sheet, sentence starters, visual graph reminders, and digital access that allows translation tools when needed. These strategies were chosen because graph interpretation in physics depends heavily on words such as slope, displacement, positive, negative, region, and transition. By reducing unnecessary language barriers, Sandra can participate more fully in the booklet, poster, and reflection tasks while still working toward the same learning goal as the class.

How this lesson shows growth in lesson planning

My lesson planning has grown most clearly in the way I now think about alignment. Earlier plans often had good activities, but this lesson shows stronger alignment among the learning goal, success criteria, assessment evidence, and the actual materials students use. The booklet, poster, observation

sheet, and 3-2-1 all now point back to the same core ideas: axes and units, slope, area, sign, and the relationship among p-t, v-t, and a-t graphs.

This lesson also shows growth in how I plan for accessibility from the start rather than treating accommodations as something added at the end. The updated packet includes scaffolded graph pages, digital options, vocabulary supports, group-role structures, and clearer checkpoints because I am getting better at anticipating where barriers may appear before students experience them. That shift matters because it makes the lesson more usable for everyone, not only for students with formal supports.

A final area of growth is assessment planning. Instead of relying too heavily on one product, I now plan for multiple forms of evidence across the lesson. The revised recording sheet reflects this growth because it captures live observation, oral explanation, collaboration, and later product evidence in one place. This makes the assessment more accurate, more fair, and more reflective of what students actually do during the activity.

Motion Graphs Student Booklet

Updated for the toy car motion investigation and graph conversions

Name: _____ **Date:** _____ **Group Members:** _____

You may show your thinking with full sentences, point form, labelled diagrams, or teacher conferencing if your teacher approves. Keep units and signs visible throughout.

Part 1 - Level Track Investigation

Objective: Collect position-time data for a toy car on a level track, graph the motion, and convert the graph using slope and area.

Role	Name(s)
Filmer / Timer	
Spotter / Release	
Recorder / Graph Lead	
Optional: Speaker / Checker	

Part 1 - Trial 1 Data

Trial 1 Data Table

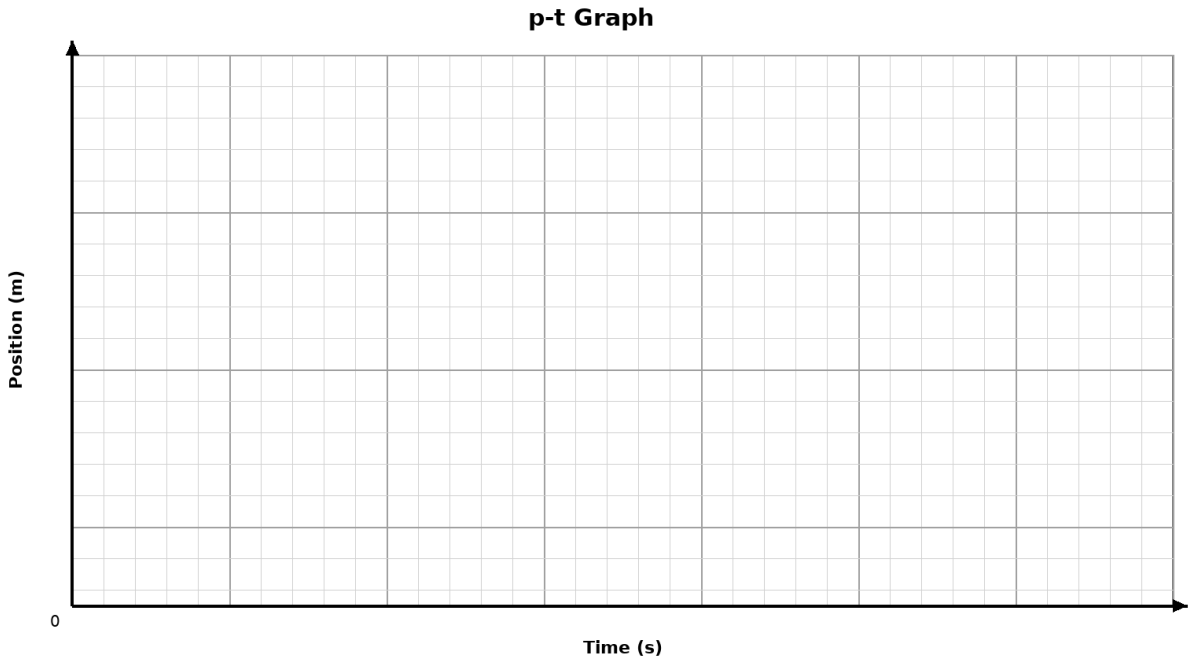
Time (s)	Position (m)

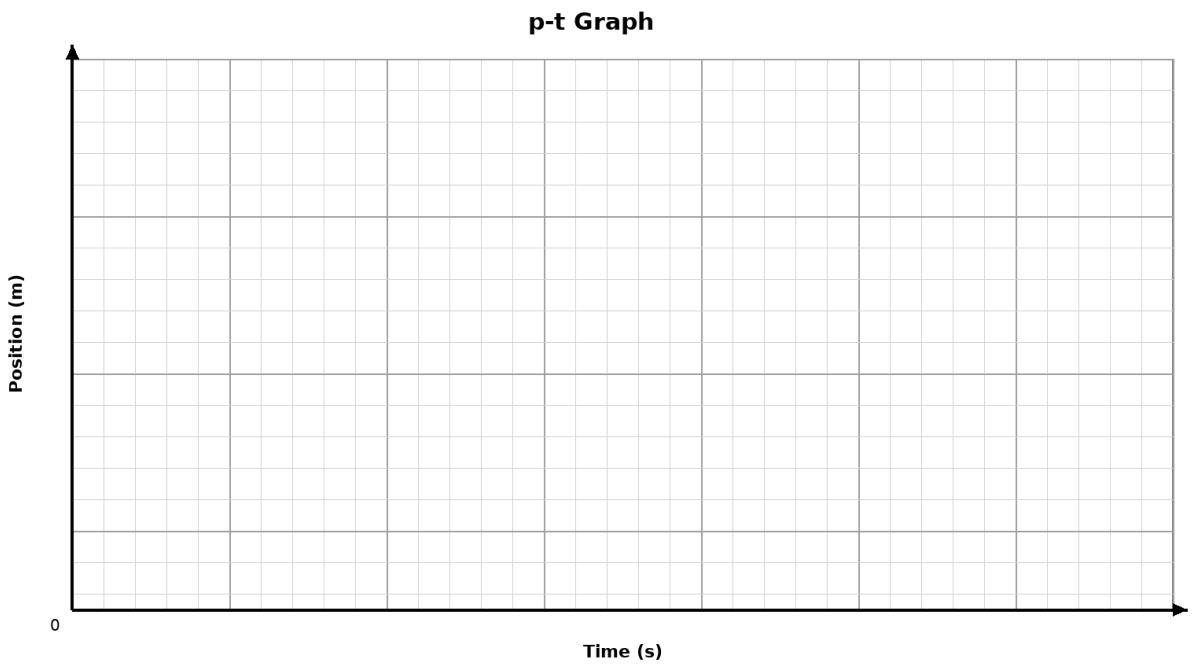
Part 1 - Trial 2 Data

Trial 2 Data Table

Time (s)	Position (m)

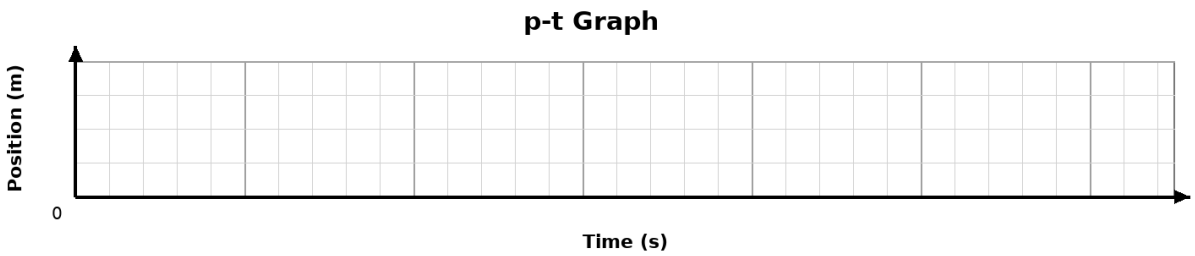
Part 1 - Trial Graphs

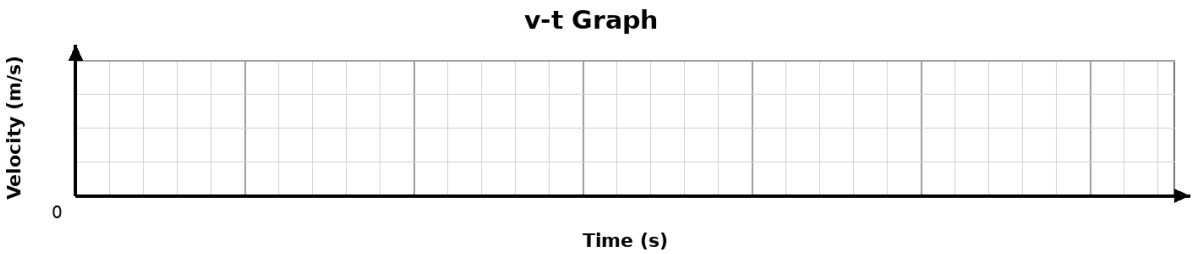


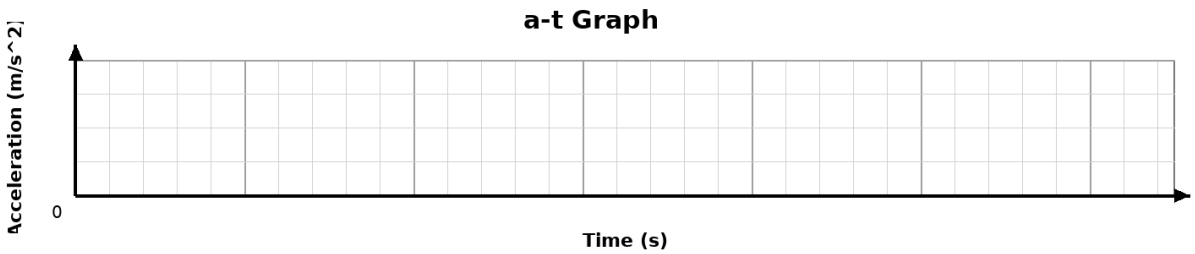


Part 1 - Convert and Explain

Choose the clearer level-track trial, draw a best-fit p-t graph, then draw the matching v-t and a-t graphs for the same time window.







Chosen points for slope on p-t Point A: _____ Point B: _____	Slope calculation and units v = Δx / Δt = _____ Units: _____

Area on your v-t graph over a chosen interval Area = _____	Cross-check on your p-t graph Δx over the same interval = _____

1. What does the slope of your p-t graph represent? What would happen to the slope if the car moved faster or slower?

2. How did Trial 1 and Trial 2 compare? Why is it useful to do more than one trial?

3. After you draw your v-t graph, what does the area under the v-t graph represent in this activity?

4. What is one source of error in this investigation?

Part 2 - Ramp to Level Investigation

Objective: Collect position-time data for a toy car that moves down a ramp and then across a level track, then connect p-t, v-t, and a-t using Regions A and B.

Role	Name(s)
Filmer / Timer	
Spotter / Release	
Recorder / Graph Lead	
Optional: Speaker / Checker	

Part 2 - Trial 1 Data

Trial 1 Data Table (Ramp to Level)

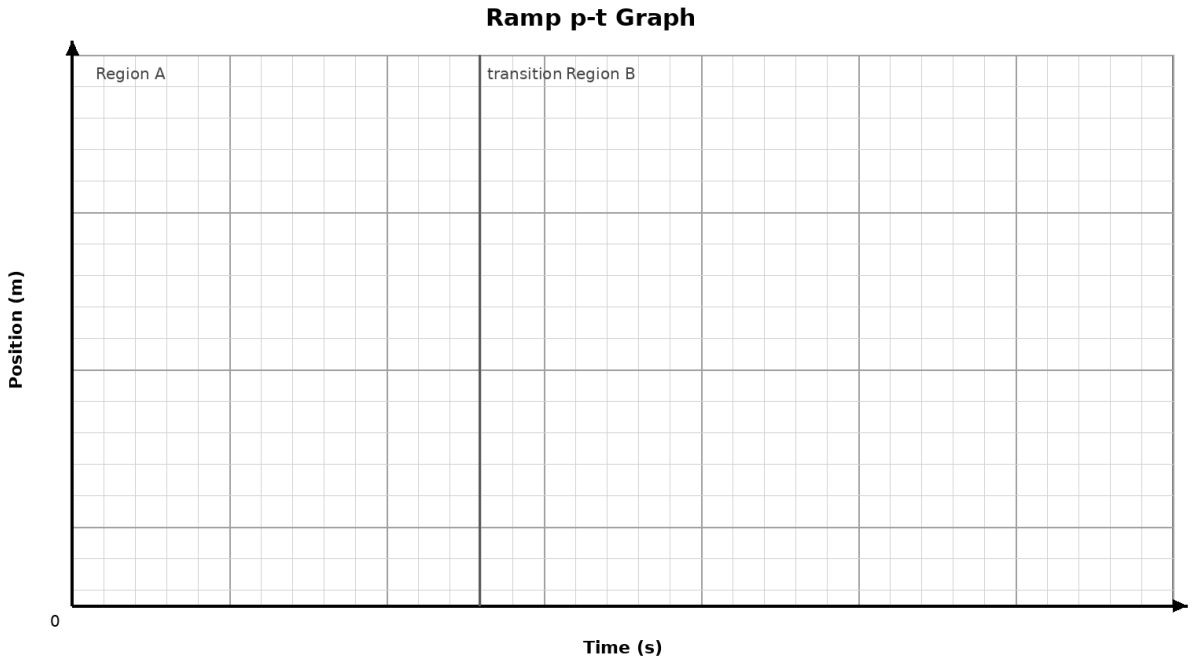
Time (s)	Position (m)

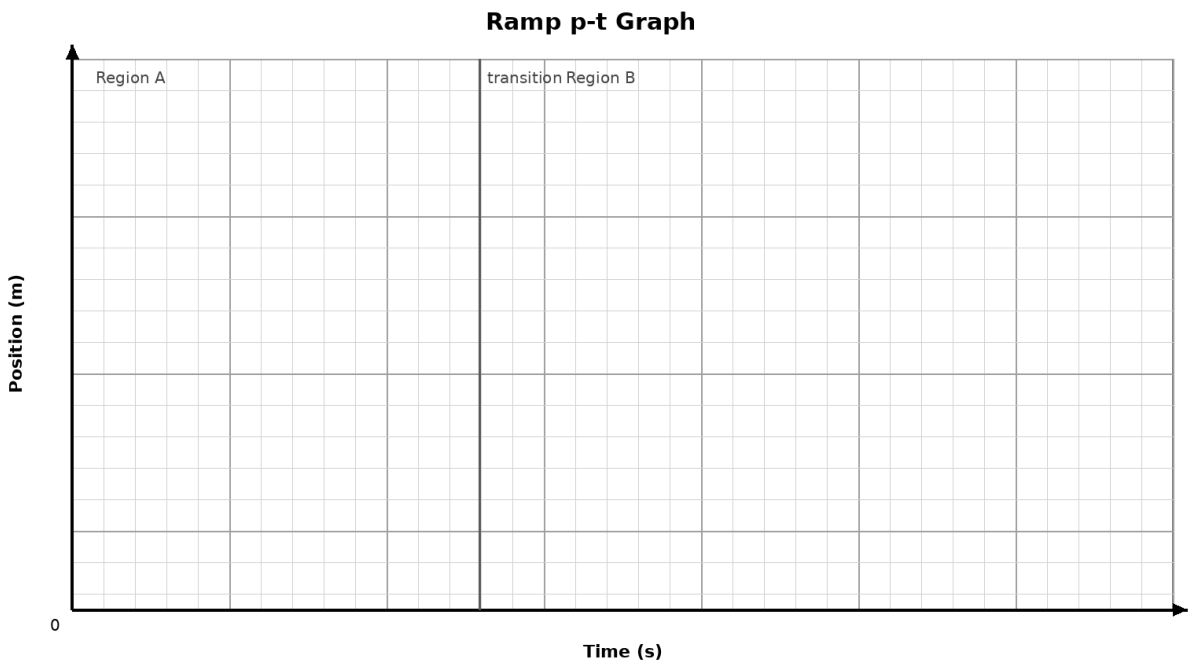
Part 2 - Trial 2 Data

Trial 2 Data Table (Ramp to Level)

Time (s)	Position (m)

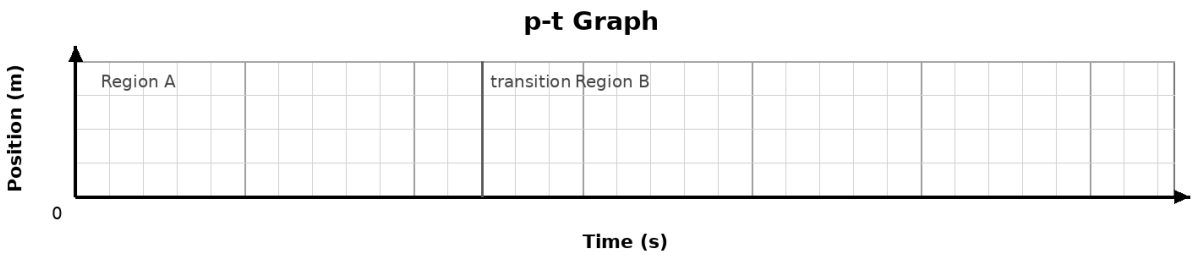
Part 2 - Trial Graphs

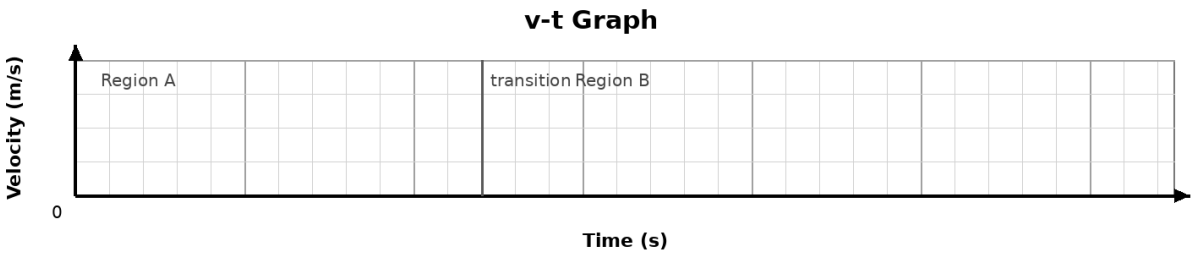


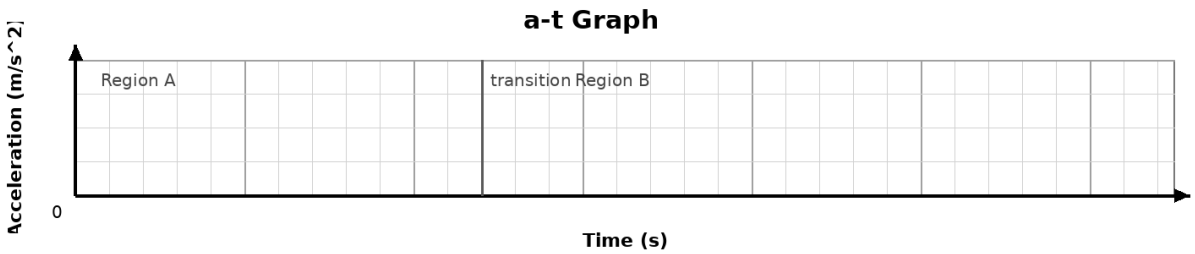


Part 2 - Convert and Explain

Choose the clearer ramp trial. Label the transition point, then draw the matching v-t and a-t graphs for the same time window.







Transition point **Ramp ends at x = _____ m** **and t = _____ s**	**Region B slope from p-t** **Choose two points and calculate the slope.** **_____**

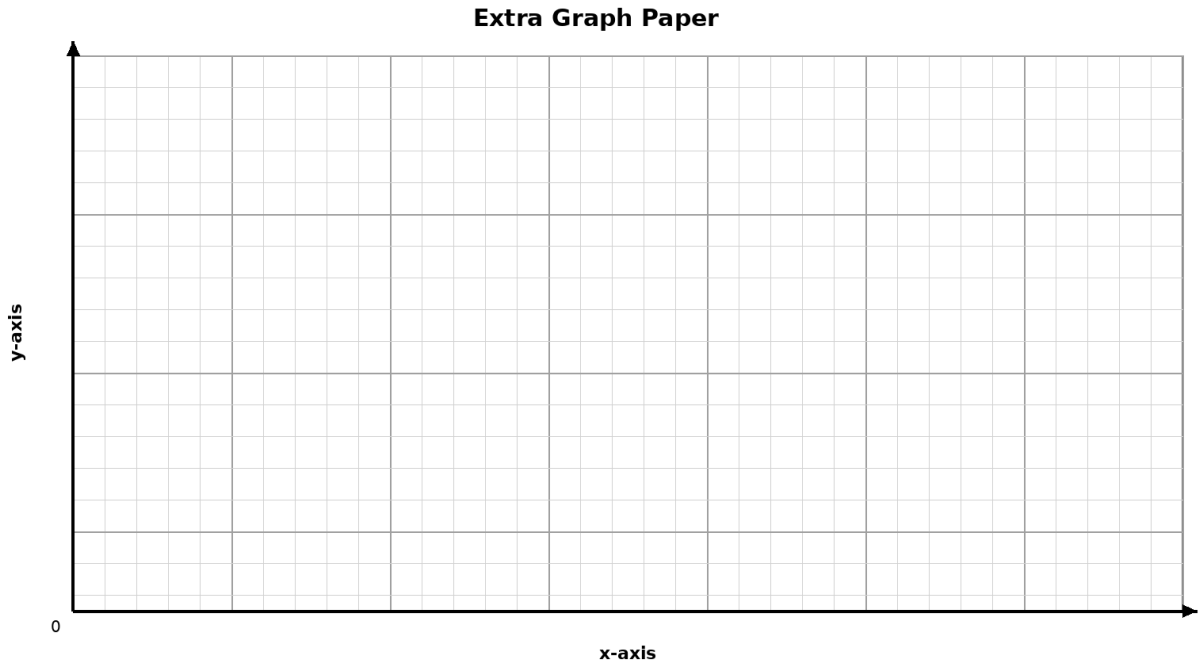
1. In one or two sentences, describe how the p-t graph changes from Region A to Region B and explain what that requires from the v-t and a-t graphs in Region B.

2. Estimate the average acceleration during Region A using your v-t graph, or state its sign and explain why.

Area under v-t over a chosen interval Predicted displacement = _____	Change on p-t over the same interval Δx = _____ Do they match? Yes / No

3. If your area check does not match perfectly, give one reason why.

Optional Extra Graph Paper



Motion Graphs Investigation Checklist

Group: _____ **Date:** _____

Use this checklist to keep your group on track. Check in with the teacher before moving on if your group is unsure.

Before we start

☐	Choose roles.
☐	Set the positive direction and origin.
☐	Make sure the device is ready and steady.
☐	Keep the metre stick or distance marks visible.

Part 1 - Level track

☐	Collect Trial 1 data.
☐	Collect Trial 2 data.
☐	Draw a best-fit p-t graph for the clearer trial.
☐	Use the p-t slope to draw the matching v-t graph.
☐	Use the v-t slope to draw the matching a-t graph.
☐	Use area on v-t to cross-check displacement on p-t.

Part 2 - Ramp to level

☐	Collect Trial 1 data.
☐	Collect Trial 2 data.
☐	Choose the clearer trial and mark the transition point.
☐	Label Region A and Region B.
☐	Draw the matching v-t and a-t graphs.
☐	Check Region B for constant velocity and zero acceleration.

Poster / final check

☐	Make sure all axes are labelled with units.
☐	Make sure the three graphs use the same time window.
☐	Include one slope statement from p-t.
☐	Include one area statement from v-t.
☐	Complete the 3-2-1 reflection.

Big 3 reminder: p-t slope = velocity | v-t slope = acceleration | area under v-t = displacement

Motion Graphs Vocabulary Support Sheet

Keep this sheet beside your booklet during the lab. Use the words, symbols, and sentence starters when you explain your graphs.

Term	Student-friendly meaning	Quick reminder / symbol
position	Where the object is compared with the origin.	x or position (m)
displacement	How much the position changed, including direction.	Δx = final - initial
velocity	How fast position changes, including direction.	m/s; from p-t slope
acceleration	How velocity changes over time.	m/s^2; from v-t slope
slope	Rise over run; change in y divided by change in x.	Δy / Δx
area under v-t	Displacement over a chosen time interval.	rectangle or triangle
positive direction	The direction chosen as positive at the start.	keep signs consistent
negative direction	Motion opposite to the chosen positive direction.	below zero / decreasing
transition point	Where the motion changes from the ramp to the level track.	label on all graphs
best-fit line	A line that shows the overall trend, not every tiny wobble.	trend vs noise

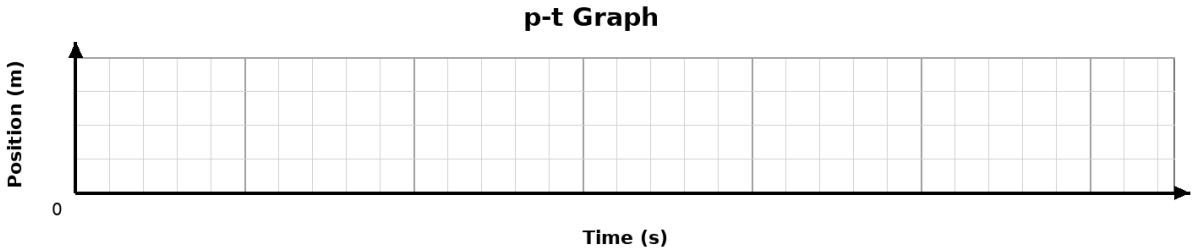
Helpful sentence starters

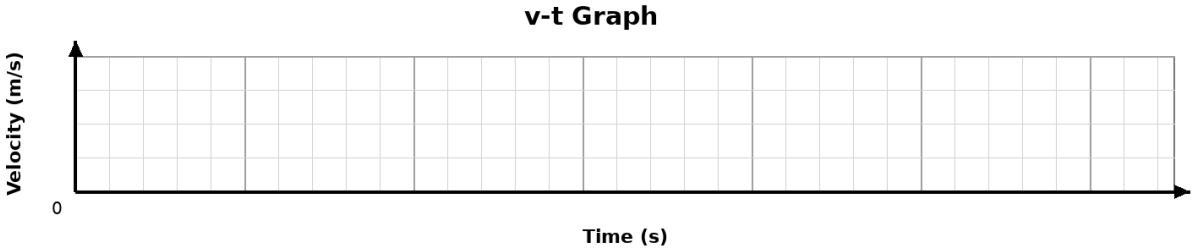
- The slope of my p-t graph shows ...
- The flat part of my v-t graph means ...
- The area under my v-t graph from t1 to t2 represents ...
- Region A shows ... because ...
- Region B shows ... because ...

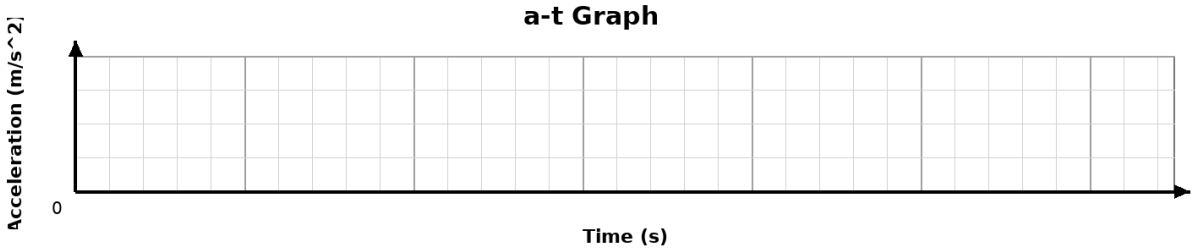
Scaffolded Graph Templates - Motion Graphs

Use these pages when students need axis labels, common graph titles, and aligned time axes already prepared.

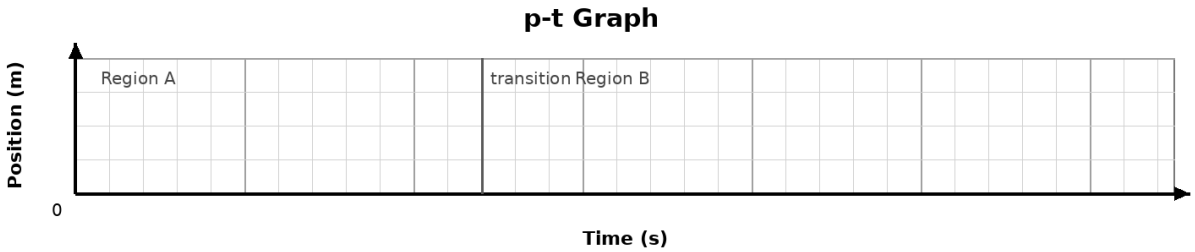
Template 1 - Standard p-t / v-t / a-t set

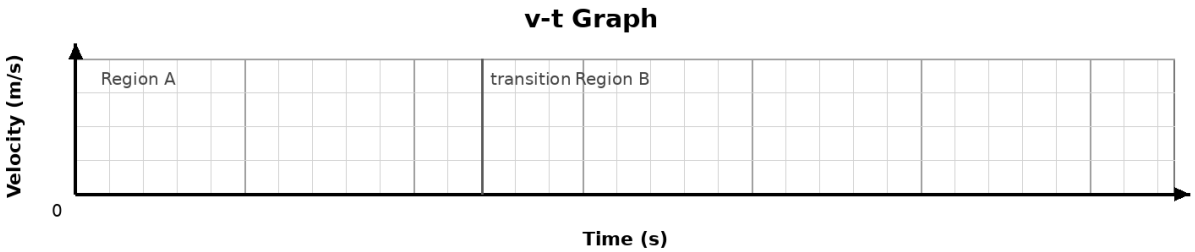


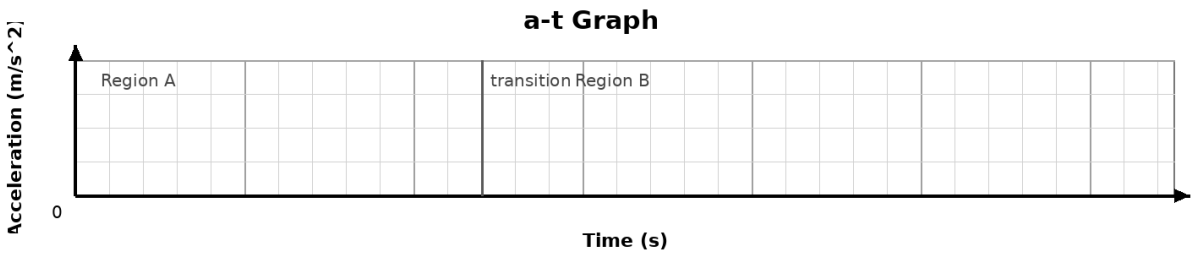




Template 2 - Ramp to level set with Region A / Region B support







Motion Graphs Group Role Cards

Cut and place one card at each group. If a group has only three members, combine Speaker / Checker with another role.

Filmer / Timer • Keep the device steady and side-on. • Record or time each trial carefully. • Make sure the track marks are visible.	**Spotter / Release** • Place the car at the correct start point. • Release the car consistently. • Watch for missed marks or unclear runs.
Recorder / Graph Lead • Fill in the table clearly. • Lead the graph setup. • Check axes, scales, units, and labels.	**Speaker / Checker** • Explain the group's thinking. • Check slope, area, and sign. • Support the poster and peer feedback.

Slo-Mo Filming Quick Guide

Use this as a posted reminder or small handout. Keep the setup simple so the filming supports the physics rather than slowing the lesson down.

Filming steps

- Place the device side-on so the full track is visible.
- Keep the device steady on a stand, book, or other stable surface.
- Make sure the distance marks and the metre stick are visible in the frame.
- Record the trial without moving the camera.
- Use slow motion or frame-by-frame scrubbing to read the time when the car passes each mark.
- Redo the trial if the car, marks, or timing are not clear.

Quick checks

- Airplane mode or notifications off.
- Positive direction set before recording.
- Use the same start point each trial.
- Keep only one person handling the device at a time.

3-2-1 Connect and Reflect - Motion Graphs

Name: _____ **Date:** _____

Keep your p-t, v-t, and a-t graphs visible. You may respond in full sentences, point form, labelled sketches, or a brief teacher conference if approved.

A) 3 Connections

Sentence starters you may use: 'The slope of my p-t graph shows ...' 'The flat part of my v-t graph means ...' 'The area under my v-t graph matches ...'

Connection 1

Connection 2

Connection 3

B) Peer Review - 2 Fixes or Glow + Next Step

Reviewer name: _____ **Fix 1 or GLOW**	**Fix 2 or GLOW**

Reviewer check: units | signs | axes labels | slope vs height | area on v-t | region labels

C) Metacognition - Choose two prompts

☐	One strategy that helped me today was ...
☐	One place I got stuck was ... and I fixed it by ...
☐	I am still wondering / not sure why ...
☐	One graph connection I can now explain more clearly is ...

Response space

Self-assessment: Understanding of today's lab 1 2 3 4 5